SOFTWARE ENGINEERING LAB

LAB 7

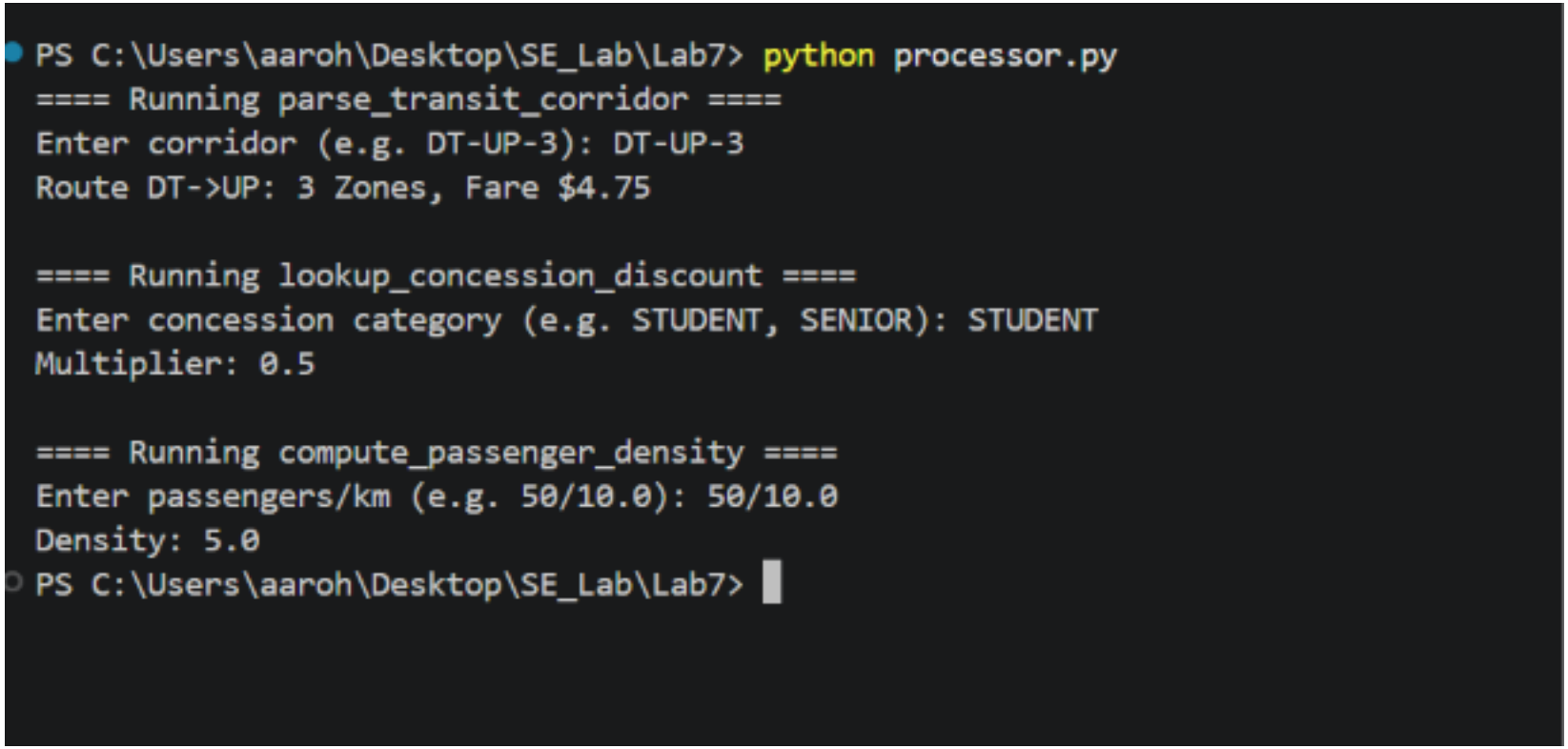
FUZZING FOR INPUT VALIDATION BUGS

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SRN: PES1UG24CS009

Section: 5A

Original Application Output



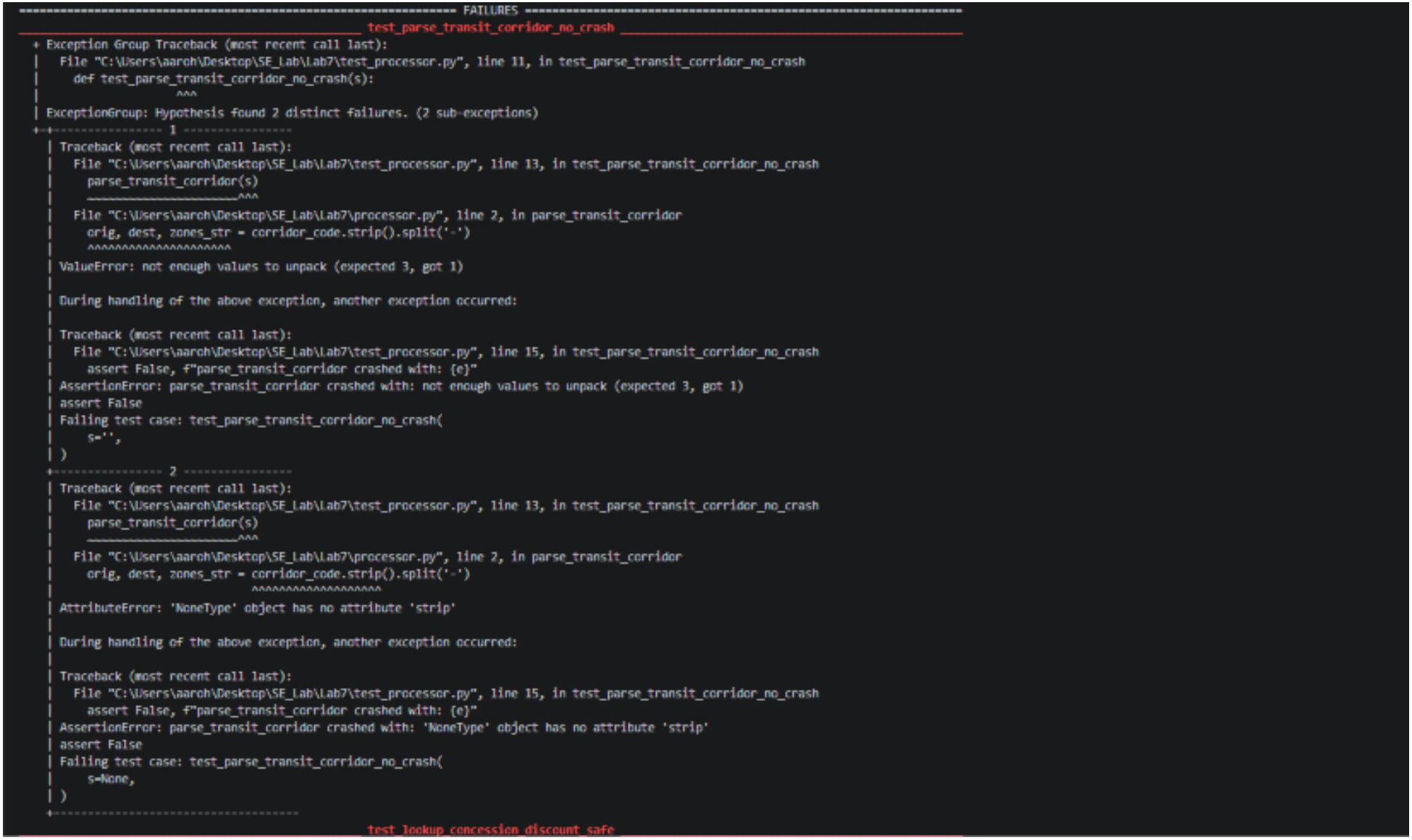
Initial Fuzz Testing Failures

```
PS C:\Users\aaroh\Desktop\SE_Lab\Lab7> python -m pytest test_processor.py -v --tb=short
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\aaroh\Desktop\SE_Lab\Lab7
plugins: anyio-4.10.0, hypothesis-6.170.14
collected 3 items

test_processor.py::test_parse_transit_corridor_no_crash FAILED                                                                      [ 33%]
test_processor.py::test_lookup_concession_discount_safe FAILED                                                                      [ 66%]
test_processor.py::test_compute_passenger_density_safe FAILED                                                                       [100%]

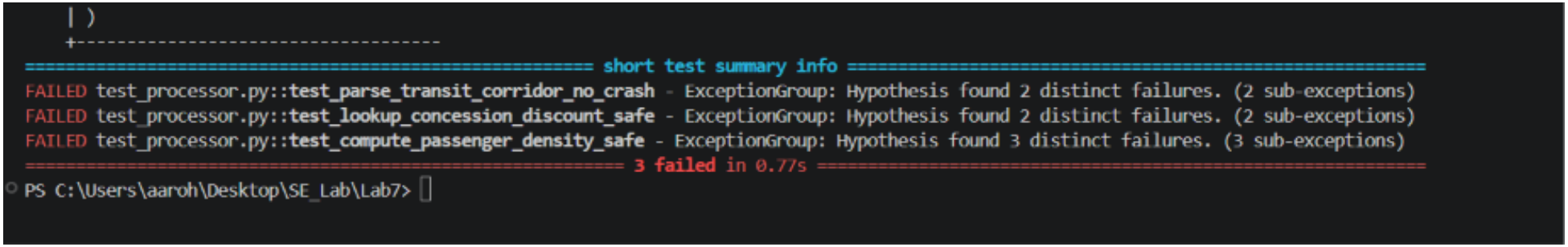
================================================================= FAILURES ==================================================================
                                                test_parse_transit_corridor_no_crash
```

Detailed Fuzz Testing Failures



```
=================================================== FAILURES ===================================================
_____________________________________ test_parse_transit_corridor_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\aarch\Desktop\SE_Lab\Lab7\test_processor.py", line 11, in test_parse_transit_corridor_no_crash
  |     def test_parse_transit_corridor_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\aarch\Desktop\SE_Lab\Lab7\test_processor.py", line 13, in test_parse_transit_corridor_no_crash
    |     parse_transit_corridor(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\aarch\Desktop\SE_Lab\Lab7\processor.py", line 2, in parse_transit_corridor
    |     orig, dest, zones_str = corridor_code.strip().split('-')
    |     ^^^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 3, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\aarch\Desktop\SE_Lab\Lab7\test_processor.py", line 15, in test_parse_transit_corridor_no_crash
    |     assert False, f"parse_transit_corridor crashed with: {e}"
    | AssertionError: parse_transit_corridor crashed with: not enough values to unpack (expected 3, got 1)
    | assert False
    | Falsing test case: test_parse_transit_corridor_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\aarch\Desktop\SE_Lab\Lab7\test_processor.py", line 13, in test_parse_transit_corridor_no_crash
    |     parse_transit_corridor(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\aarch\Desktop\SE_Lab\Lab7\processor.py", line 2, in parse_transit_corridor
    |     orig, dest, zones_str = corridor_code.strip().split('-')
    |                             ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\aarch\Desktop\SE_Lab\Lab7\test_processor.py", line 15, in test_parse_transit_corridor_no_crash
    |     assert False, f"parse_transit_corridor crashed with: {e}"
    | AssertionError: parse_transit_corridor crashed with: 'NoneType' object has no attribute 'strip'
    | assert False
    | Falsing test case: test_parse_transit_corridor_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________________ test_lookup_concession_discount_safe _____________________________________
```

Fuzz Testing Failure Summary



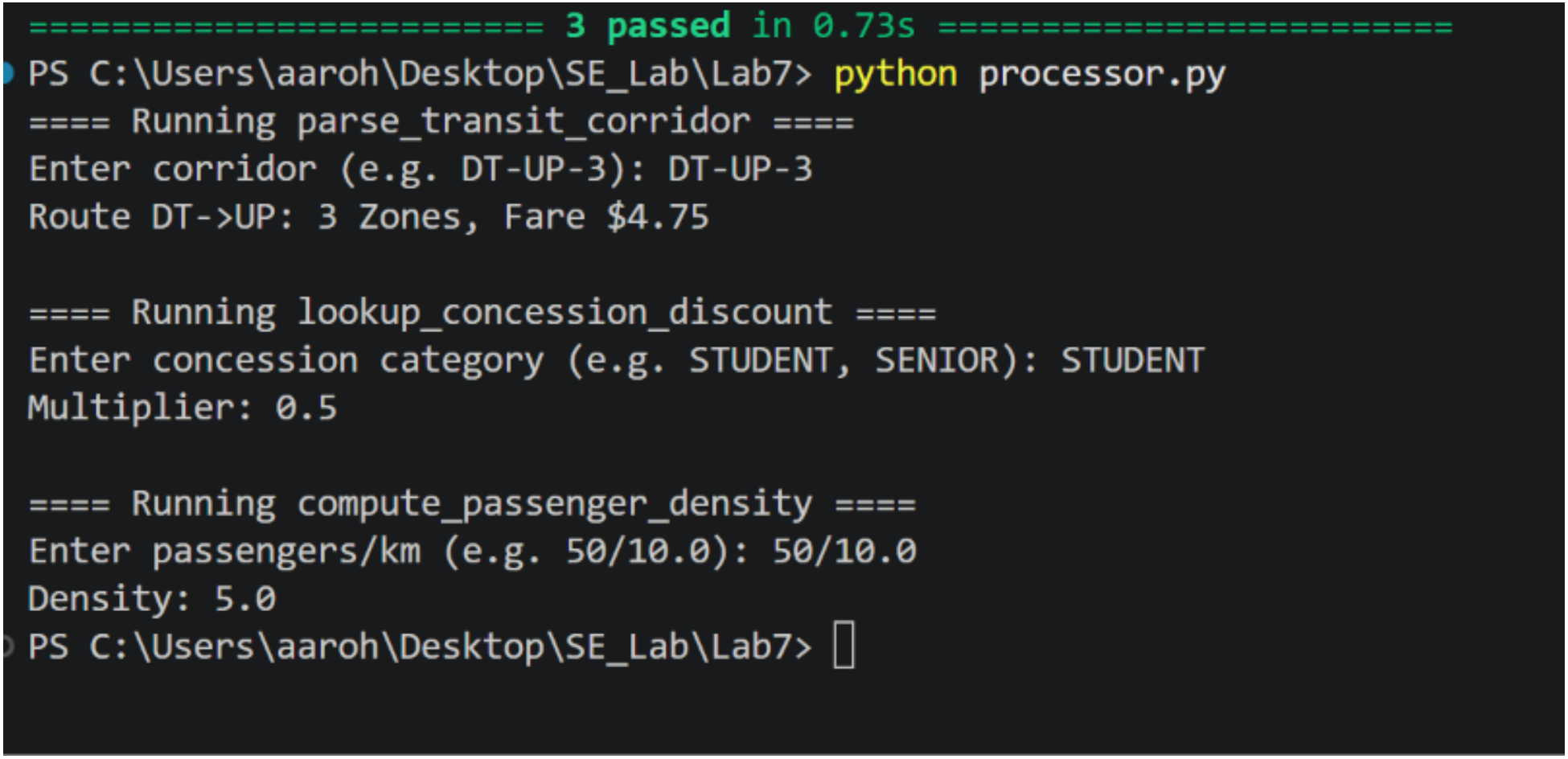
Fuzz Tests Passing

```
PS C:\Users\aaroh\Desktop\SE_Lab\Lab7> python -m pytest test_processor.py -v
========================= test session starts =========================
platform win32 -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python313\python.exe
cachedir: .pytest_cache
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plugins: anyio-4.10.0, hypothesis-6.170.14
collected 3 items

test_processor.py::test_parse_transit_corridor_no_crash PASSED [ 33%]
test_processor.py::test_lookup_concession_discount_safe PASSED [ 66%]
test_processor.py::test_compute_passenger_density_safe PASSED  [100%]

========================== 3 passed in 0.73s ==========================
PS C:\Users\aaroh\Desktop\SE_Lab\Lab7>
```

Fixed Application Output



Final Fuzz Tests Passing

